Null Choir

Reverse Engineering



I began by triaging the provided files: stripped ELF64 binary null_choir, 196 KB bytecode blob canticle.score (magic NCS2), and trace blob echoes.trace (magic ECH8).

Reverse engineering revealed a custom register VM interpreter that maps a 48-byte input across six 64-bit registers, executes 4,096 instructions, and verifies the state against an encrypted target.

I authenticated the 32 genuine records from echoes.trace to recover the conductor seed, identified instruction bank 0, and reversed all 4,096 VM instructions and diffusion layers backwards to invert the initial state.

Finally, I tested the recovered flag against the executable / service:

./null_choir 'SecLeaf{th3_ch01r_s1ng5_1n_5h4d0w_m4ch1n35!!!!!}'

The program returned Harmony, confirming that the flag was correct.

Flag Captured

SecLeaf{th3_ch01r_s1ng5_1n_5h4d0w_m4ch1n35!!!!!}

BY:JOKER_53